

Learned-Energy Particle Beliefs for Continuous POMDPs: Experimental Comparison

Results snapshot, 22 September 2026

1 Setup

Training and evaluation protocol. Every policy is trained with PPO for 200 updates. Each update collects 128 steps from 16 parallel environments (409,600 environment steps in total), uses 4 epochs of minibatch size 512, learning rate 2.5×10^{-4} , discount 0.99, GAE $\lambda = 0.95$, and a symlog transform of the value targets. Every 25 updates the policy is evaluated on 10 held-out episodes with a deterministic decoding of the action distribution; the tables report the undiscounted return averaged over those episodes. Each configuration is trained with five seeds (10–14). The seed fixes the network initialisation, the training-environment streams and the held-out evaluation episodes, so the same seed corresponds to the same evaluation set for every method.

Proposed method: particle beliefs driven by a learned energy. The agent keeps a belief as a set of $K = 16$ particles in a learned latent space. At each time step the particles are moved by $L = 4$ unadjusted Langevin steps on an energy network conditioned on the current observation and previous action (step size 0.02, temperature 1). The particle set is then read by an encoder that produces the input of a diffusion policy head and the value head. Three encoders are compared:

- **Transformer encoder:** one self-attention layer over the particles ($d_{\text{model}} = 64$, 4 heads), 75,286 parameters in total.
- **α -vector head:** per-particle trunk features are mean-pooled and read by 16 linear pieces combined with a tempered log-sum-exp, i.e. a learned piecewise-linear convex functional of the belief in the spirit of α -vector value representations, 35,525 parameters. A *parameter-matched* variant widens its trunk (feed-forward width 704) to 75,269 parameters.
- **DeepSets encoder:** a permutation-invariant sum-pooling network, 67,222 parameters.

Four belief-update designs are studied. They differ in how the particles are initialised at each step and in whether auxiliary prediction losses are added to the policy-gradient objective:

- **Design A:** particles are re-drawn from $\mathcal{N}(0, I)$ at every step before the Langevin moves; no auxiliary loss.
- **Design B:** particles persist from the previous step (warm start), and the first particle set of an episode is drawn from a learned observation-conditioned proposal $\mu(o_0) + \sigma(o_0) \xi$ instead of $\mathcal{N}(0, I)$; auxiliary losses predict the reward and the next observation from the belief (coefficient 1 each).
- **Design C:** re-drawn particles as in A, with the auxiliary losses of B.
- **Design D:** persistent particles and the learned initial proposal as in B, without auxiliary losses.

Baselines. A GRU policy (75,674 parameters), a plain RNN policy (75,970 parameters), and a particle-filter policy in the style of PF-RNN/DPFRL with a learned stochastic transition, learned observation weights and soft resampling at every step, sized to the same parameter budget (76,080 parameters). For Mountain Hike an earlier particle-filter implementation with deterministic resampling is also listed for seeds 10–12.

Tasks. *Mountain Hike* (Igl et al., 2018): noisy 2-D terrain navigation. The agent starts near one corner of a box, moves by a norm-clipped step of at most 0.5 with Gaussian transition noise (standard deviation 0.25), observes its position with Gaussian noise of standard deviation 3, and receives the terrain height as reward at every step minus an action cost, with a flat penalty outside the box; episodes last 75 steps. Three scripted reference policies calibrate the scale: a stationary policy obtains -360 , a policy that reacts to the raw observation only obtains -189 , and a policy with access to the true position obtains -122 . *Light-Dark 2D*: the classical planar light-dark navigation task with a 30-step horizon. The state follows $x' = x + u$, the observation noise grows with the distance from the light at $x_1 = 5$, the agent starts around $x_1 = 2$ (standard deviation 1), pays a small quadratic action cost, and pays a terminal cost (weight 10) for its final distance from the goal at the origin. No scripted references have been computed for this task; earlier single-seed runs of the same task reached -25 (Transformer encoder) and -29 (GRU policy).

2 Results

Table 1: Mountain Hike: evaluation return after 200 updates (deterministic policy, 10 held-out episodes) for each training seed. Designs A–D are defined in Section ?? . *: run stopped at update 175 (wall-time limit), latest evaluation shown. The last column averages completed seeds only.

Design	Policy	Seed 10	Seed 11	Seed 12	Seed 13	Seed 14	Mean $\pm$ SD
A	Transformer encoder	−443	−445	−182	−447	−263	−356 $\pm$ 125
	α -vector head	−270	−291	−277	−280	−286	−281 $\pm$ 8
	α -vector head (parameter-matched)	−270	−447	−278	−293	−295	−317 $\pm$ 66
	DeepSets encoder	−430	−284	−442	−447	−283	−377 $\pm$ 86
B	Transformer encoder	−129	−439	−446	−381	−222	−323 $\pm$ 141
	α -vector head	−213	−282	−143	−268	−240	−229 $\pm$ 55
	α -vector head (parameter-matched)	−198	−436	−250	−183	−232	−260 $\pm$ 91
	DeepSets encoder	−200	−236	−447	−254	−437	−315 $\pm$ 118
C	Transformer encoder	−442	−253	−447	−442	−255	−368 $\pm$ 104
	α -vector head	−258	−288	−290	−277	−285	−280 $\pm$ 13
	α -vector head (parameter-matched)	−448*	−196	−335*	−447*	−258*	–
	DeepSets encoder	−447	−273	−284	−447	−263	−343 $\pm$ 95
D	Transformer encoder	−277	−285	−276	−279	−227	−269 $\pm$ 24
	α -vector head	−278	−182	−273	−281	−278	−258 $\pm$ 43
	α -vector head (parameter-matched)	−267	−274	−284	−269	−275	−274 $\pm$ 6
	DeepSets encoder	−205	−250	−262	−285	−261	−253 $\pm$ 30
Baselines	GRU policy	−115	−153	−168	−279	−225	−188 $\pm$ 65
	RNN policy	−128	−162	−132	−202	−277	−180 $\pm$ 62
	Particle-filter policy	−165	−162	−129	−146	−126	−146 $\pm$ 18
	Particle-filter policy (earlier version)	−447	−201	−279	–	–	−309 $\pm$ 126

Table 2: Light-Dark 2D: evaluation return after 200 updates (deterministic policy, 10 held-out episodes) for each training seed. * as above; “diverged” marks a run that stopped with a non-finite gradient.

Design	Policy	Seed 10	Seed 11	Seed 12	Seed 13	Seed 14	Mean $\pm$ SD
A	Transformer encoder	-29	-161	-94	-46	-62	-78 $\pm$ 52
	α -vector head	-158	-179	-378	-91	-125	-186 $\pm$ 112
	α -vector head (parameter-matched)	-313	-133	-394	-25	-285	-230 $\pm$ 133
	DeepSets encoder	-29	-91	-63	-70	-48	-60 $\pm$ 23
B	Transformer encoder	-39	-294	-61	-46	-25	-93 $\pm$ 113
	α -vector head	-85	-304	-226	-44	-196	-171 $\pm$ 106
	α -vector head (parameter-matched)	-3956	-464	-4998	-189	-450	-2011 $\pm$ 2043
	DeepSets encoder	-55	-69	-79	-33	-68	-61 $\pm$ 18
C	Transformer encoder	-48	-120	-300	-102	-239	-162 $\pm$ 104
	α -vector head	-51	-565	-217	-468	-292	-318 $\pm$ 204
	α -vector head (parameter-matched)	-41	-95	-111	-54	-1043	-269 $\pm$ 388
	DeepSets encoder	-23	-209	-96	-371	-140	-168 $\pm$ 132
D	Transformer encoder	-47	-119	-94	-75	-173	-102 $\pm$ 48
	α -vector head	-106	-437	-168	-101	-373	-237 $\pm$ 157
	α -vector head (parameter-matched)	-307	-211	-74	-202	-163	-191 $\pm$ 75
	DeepSets encoder	-63	-125	-246	-43	-141	-124 $\pm$ 80
Baselines	GRU policy	-29	-104	-70	-69	-70	-68 $\pm$ 27
	RNN policy	-74	-59	-92	-65	-50	-68 $\pm$ 16
	Particle-filter policy	diverged	-53	-72	-55	-57	-59 $\pm$ 9

Table 3: Light-Dark 2D: mean of the last four evaluations (updates 125–200) for each training seed. This damps the evaluation-to-evaluation volatility of the α -vector runs and gives a more stable ranking than the single update-200 value.

Design	Policy	Seed 10	Seed 11	Seed 12	Seed 13	Seed 14	Mean $\pm$ SD
A	Transformer encoder	-218	-89	-67	-58	-79	-102 $\pm$ 66
	α -vector head	-266	-166	-201	-322	-207	-232 $\pm$ 62
	α -vector head (parameter-matched)	-434	-411	-318	-607	-466	-447 $\pm$ 94
	DeepSets encoder	-228	-57	-116	-72	-45	-104 $\pm$ 75
B	Transformer encoder	-110	-181	-260	-126	-134	-162 $\pm$ 60
	α -vector head	-152	-305	-150	-230	-563	-280 $\pm$ 171
	α -vector head (parameter-matched)	-2495	-867	-1452	-735	-635	-1237 $\pm$ 690
	DeepSets encoder	-136	-90	-53	-202	-227	-142 $\pm$ 73
C	Transformer encoder	-362	-130	-156	-71	-216	-187 $\pm$ 111
	α -vector head	-213	-246	-89	-359	-359	-253 $\pm$ 113
	α -vector head (parameter-matched)	-283	-213	-575	-105	-426	-320 $\pm$ 165
	DeepSets encoder	-199	-176	-86	-419	-220	-220 $\pm$ 122
D	Transformer encoder	-127	-117	-100	-64	-149	-111 $\pm$ 32
	α -vector head	-272	-429	-142	-365	-290	-300 $\pm$ 108
	α -vector head (parameter-matched)	-396	-270	-356	-1272	-332	-525 $\pm$ 376
	DeepSets encoder	-154	-117	-229	-168	-190	-172 $\pm$ 42
Baselines	GRU policy	-41	-82	-57	-54	-85	-64 $\pm$ 19
	RNN policy	-78	-55	-70	-47	-57	-62 $\pm$ 12
	Particle-filter policy	diverged	-58	-77	-51	-65	-63 $\pm$ 11

3 Observations

- **Mountain Hike.** No belief design reaches the recurrent or particle-filter baselines (-146 to -188 on average) or the observation-only reference (-189). Persistent particles (Designs B and D) avoid the degenerate policy that scores about -445 more often than re-drawn particles (A and C). The α -vector head is the most stable encoder (-258 to -281 across designs, standard deviation below 55 in every design), but it also plateaus there. Auxiliary prediction losses on their own (Design C) do not prevent the degenerate policy. The single strong Transformer run in Design B (seed 10, -129) is not reproduced by the other four seeds.
- **Light-Dark 2D.** The DeepSets encoder in Designs A and B (-60 and -61 on average) matches the baselines (-59 to -68). The Transformer encoder is competitive except for one collapsed seed per design. The α -vector head is worse on average, and its evaluation return swings between about -25 and -1500 from one evaluation to the next, so a single update-200 value is largely a sampling artefact; Table ?? averages the last four evaluations to damp this.
- **Parameter-matched α -vector head.** Widening the α -vector trunk to the Transformer’s parameter budget does not help: on Mountain Hike it pushes Designs A and C into the degenerate policy on most seeds, and on Light-Dark 2D it makes the evaluations even more volatile. On Mountain Hike four Design-C seeds stopped at update 175 (wall-time limit); asterisks mark their latest evaluation.
- **Seed effects.** Because all methods share the held-out episodes of a given seed, per-seed columns are directly comparable across rows. The held-out sets of the five seeds are of similar difficulty (their across-method mean returns differ by less than 10 on both tasks), so the differences between columns reflect training variability rather than evaluation difficulty. On Light-Dark 2D the apparent advantage of seed 10 in the update-200 table disappears in the last-four-evaluation average.
- **Divergence.** One particle-filter run on Light-Dark 2D (seed 10) stopped at update 46 with a non-finite gradient norm; resuming from its checkpoint reproduces the failure because the stored rollout is replayed deterministically.